Ashwin Chembu

(341) 356-1722 | ashchembu@gmail.com | San Francisco, California

PROFILE

Backend engineer building Python and Node.js APIs, storage integrations, authentication, and event-driven workflows. Focuses on secure service boundaries, durable data handoffs, and repeatable validation.

TECHNICAL SKILLS

Languages / APIs: Python, TypeScript, JavaScript, SQL, FastAPI, Flask, Node.js, Express, OpenAPI, MCP
Storage / Cloud: MongoDB, Redis, AWS S3, Firebase, Render, WebSockets
Reliability: HMAC webhooks, authentication, validation, rate limiting, idempotency, pytest

EXPERIENCE

Business Technology Solutions Associate | ZS Associates — **South San Francisco, CA | Jun 2024-Present**

- Built **Python**/Flask endpoints connecting AWS S3 and Smartsheet APIs, coordinating document storage and workflow handoffs so teams could ingest, organize, and evaluate files through integrated cloud services.
- Built a **Node.js** provider-lookup workbench with exact identifier search, Jaro-Winkler address ranking, bulk uploads, and session-bound exports; validated authentication, isolation, input checks, and export safety with **10** local tests.
- Built React/FastAPI RAG assistants with AWS document ingestion and Redis-backed services, connecting uploaded knowledge to searchable enterprise workflows so teams could retrieve context through usable product interfaces.
- Defined layered ETL/MDM architecture from landing and staging through transformation, canonical records, and publication, with data-quality checks, delta logic, and history tracking to preserve traceable changes between sources.
- Added cached-section recovery and explicit retries to React CRM views, preserving last-good data after refresh failures and ignoring obsolete requests; **five** regression tests covered initial errors, retries, and selected-plan changes.

Founder and Lead Developer | Tallyrus — **Jun 2023-Sep 2025**

- Implemented a Flask grading API that validates rubric, essay, prompt, and historical-example inputs before generating JSON feedback, connecting model requests to the review workflow without treating automated output as a final grade.
- Connected **Express**/Multer upload routes for teacher files, rubrics, and historical essays to AWS S3 storage and **MongoDB** assignment records, making educator materials available to downstream grading and review workflows.
- Integrated React/TypeScript and **Express** product workflows with Google OAuth, protected routes, **MongoDB** persistence, and secure sessions, connecting authenticated educator access to document and essay-feedback operations.
- Integrated LangChain/OpenAI embeddings with NumPy normalization and FAISS similarity search, combining retrieved historical essays with teacher rubrics and prompts in GPT-4o requests to return contextual feedback for educator review.
- Integrated a pretrained Hugging Face Transformers classifier with PyTorch, adding sentence chunking, CPU/CUDA selection, and length-weighted prediction aggregation behind a Flask API to handle longer text inputs.

Data Analyst / Web Development Lead | Tech4Good at UCSC — **Mar 2023-Jun 2025**

- Implemented **20+** attributed Node/**Express** REST routes with **MongoDB** persistence and logging for chat, text-entry, and polling workflows, supporting threaded messages, anonymous responses, editing, voting, and admin controls.
- Separated chat, text-entry, and polling state across **three MongoDB databases**, using change streams, populated message logs, and persisted voter records to synchronize threads, edits, group responses, and reveal/reset operations.
- Built TypeScript/Figma API chat, text-entry, polling, and Excel/XLSX export workflows in a collaborator-owned FigJam repository, enabling research and classroom teams to collect, discuss, and export structured participant responses.
- Analyzed **43,340** anonymized interaction rows across **2,805** sessions with **Python**/Pandas, calculating group active-time measures and building Excel/XLSX exports to turn collaboration records into structured research evidence.

PROJECTS

Private Filedrop | Secure File Exchange — **Jun 2026-Aug 2026**

- Built password-protected browser uploads and API-key-protected **Express** APIs with **OpenAPI** documentation, organizing files by metadata, tags, project paths, and version history so devices and AI clients could exchange artifacts securely.
- Added local and S3-compatible storage drivers with migration tooling that preserves file bytes and metadata, retaining rollback sources and unguessable download links to support durable, versioned handoffs across computers.
- Exposed **MCP** tools for listing, uploading, organizing, and retrieving files, with HMAC-SHA256 webhooks and timing-safe credential checks; added deduplicated jobs, atomic claims, renewable leases, and retry limits for cross-device coordination.

Forge | Personal Fitness Intelligence — **Personal project**

- Integrated Hevy/Oura APIs and Apple Health webhooks with FastAPI/**MongoDB,** combining workout, nutrition, sleep, and recovery records while preferring Oura fields and using Apple Health values to fill missing recovery observations.
- Implemented configurable scheduled Hevy/Oura API pulls, on-demand refresh, MyFitnessPal CSV ingestion, and authorization-reconnection handling, keeping source updates separate from Apple Health push events in the fitness data pipeline.
- Developed an analysis engine that compares actual performance with a **12-week** hypertrophy plan, scoring workout adherence, nutrition compliance, and recovery quality to turn unified fitness records into useful program summaries.

Lume | AI Meeting Intelligence — **Dec 2025-Jan 2026**

- Built a React and Node/**Express** meeting-intelligence platform with **MongoDB** records and AWS S3 file storage, connecting meeting capture, uploaded documents, transcript-aware search, and project questions in one reusable workflow.
- Designed a RAG engine with vector retrieval, intent routing, evidence extraction, and generator/critic refinement, combining relevant meeting and file context into citation-backed answers that users can trace to supporting sources.
- Built reusable Project Brain knowledge bases from transcripts and seed files, retaining project terminology, relationships, and contextual evidence so later questions could reuse accumulated knowledge across meetings and documents.

EDUCATION

University of California, Davis | B.S. Computer Science | GPA: 3.7 — **Davis, CA | Graduated Jun 2025**

LINKS

Portfolio: https://ashchembu.com | LinkedIn: linkedin.com/in/ashwin-chembu-a5292b205 | GitHub: github.com/ashwinchembu